

Sam Rutherford

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Profile

- Cybersecurity Graduate (First Class honours) from the “Ethical Hacking” course at Abertay University, renowned for its full cybersecurity focus and practical skills bias.
- Seeking a graduate / entry-level cyber-security or software engineering role to build on my strong foundational skills in coding, cyber defence, threat analysis and digital forensics.

Education

ETHICAL HACKING 1ST HONS | 2020-2024 | UNIVERSITY OF ABERTAY DUNDEE

- Degree subject: Over 4 academic years, developed an incredibly strong foundation for cybersecurity, networking, team-based software development and penetration testing.
- Honours modules:
 - Year 3: Professional project planning, Advanced network security, Web application hacking, Professional project delivery, Advanced ethical hacking, Android development.
 - Year 4: IoT security and fundamentals, Advanced digital forensics, Engineering resilient systems (secure software design, machine learning fundamentals, human centred security), Honours project.
- Honours project:
 - “Automating the detection, removal, and prevention of Mirai infection in IoT devices”. The project involved performing a malware analysis on the malware known as Mirai which specifically targets low level Linux IoT devices. Based upon this and literature review, a tool was developed which would automatically identify IoT devices in a network, assess the vulnerability of the device to Mirai, perform device hardening and detect and remove Mirai if found within the device. The results of this project were written up in a dissertation which can be provided upon request.
- Miscellaneous projects
 - Conducted various penetration tests utilising Linux based tools.
 - Performed numerous malware analysis.
 - Performed various digital and network forensics investigations.
 - Developed various networking tools using SCAPY.
 - Acted as project manager for the development of a visualised network monitoring tool.
 - Used AI tools to develop a red team phishing tool.
 - Extensive report writing performed for above projects.

ABERDEEN GRAMMAR SCHOOL (2014-2020), ABERDEEN SCOTLAND

- Advanced Highers: Maths, Physics, Chemistry
- Highers: Maths, Physics, Chemistry, English, Computing Science.

Skills & Abilities

- Software development
- Familiar with Waterfall, Agile, Prototype and Scrum
- Adept in various programming languages

- HTML, CSS, C++, C, Python, Javascript
- Adept with hacking tools
- NMAP, burpsuite, Nessus, Metasploit, SQLMap, Wireshark, Ghidra
- Excellent report writing and presentation skills
- Digital and network forensics skills
- Malware analysis and development including process injection
- CTF experience, including pwn box
- Familiar with using AI tools
- Team-based project planning and delivery
- Additionally, experience as project lead
- Familiar with using AWS
- Amplify, Lambda and boto3

Experience

WORK EXPERIENCE | BP | 2017

- Varied office work experience assisting engineering support team for oil and gas operation. Work included data verification, database development and conducting meetings to identify user requirements.

Extracurricular activities

“CYBER 9/12 CHALLENGE” (ATLANTIC COUNCIL), 2023

- Participated in the “Cyber 9/12” challenge in a team of 4 and acted as the team leader. The tasks included creating rules and a suitable incident response to cyber-attacks and risks in a strict timeframe and presenting them to a board which provided valuable feedback for our talk and policies. Our team came 4th out of 17 teams and gained valuable knowledge of problem solving as a team, working under pressure and presenting to clients.

PRESENTED AT SECURITAY, 2024

- Presented at SecuriTay which is the largest student ran cybersecurity conference in the UK. The presentation covered the history, examples and how phishing is used within the modern day. Additionally, a previously self-developed AI phishing tool was showcased displaying the AI capabilities for red team developments and how AI can still be used for developing countermeasures.

HACKTHEBOX, 2023-2025

- Participated in many hacking challenges on the website HackTheBox to further my understanding of using hacking tools. These challenges, often known as ‘CTFs’ greatly improve and refresh current hacking skills.

VOLUNTEER, CANCER RESEARCH UK

- Worked behind the counter, setting up stock, processing donated items and taking in deliveries. Refined skills of teamwork and communication which came from meeting a new team and being welcoming and friendly towards customers.